H1B VISA APPLICATION ANALYSIS (2025)

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OBJECTIVE

-Analyze H1B visa application trends

-Identify valid vs total applications

-Evaluate approval rate and case status distribution

-Discover top employers, job roles, and salary trends

-Analyze state-wise hiring patterns

DATASET OVERVIEW

- Source: Public H1B Visa Applications Dataset
- Total Records: ~2.4 Million (Raw Data)
- Valid Applications: ~596K (After Data Cleaning & Removing Duplicates)
- Key Features:
- -Case Status (Certified, Denied, Withdrawn)
- -Employer Name
- -Job Title
- -Salary
- -Work Location (State)
- -Application Date

TOOL USED

- • SQL → Data analysis
- • Power BI → Dashboard

KEY PERFORMANCE INDICATORS

- Total Valid Applications analyzed:596K+
- Number of Employers involved:72K+
- Average Salary offered: $389K
- Unique Job Roles identified: 120K+
- Overall Approval Rate: 92.5%

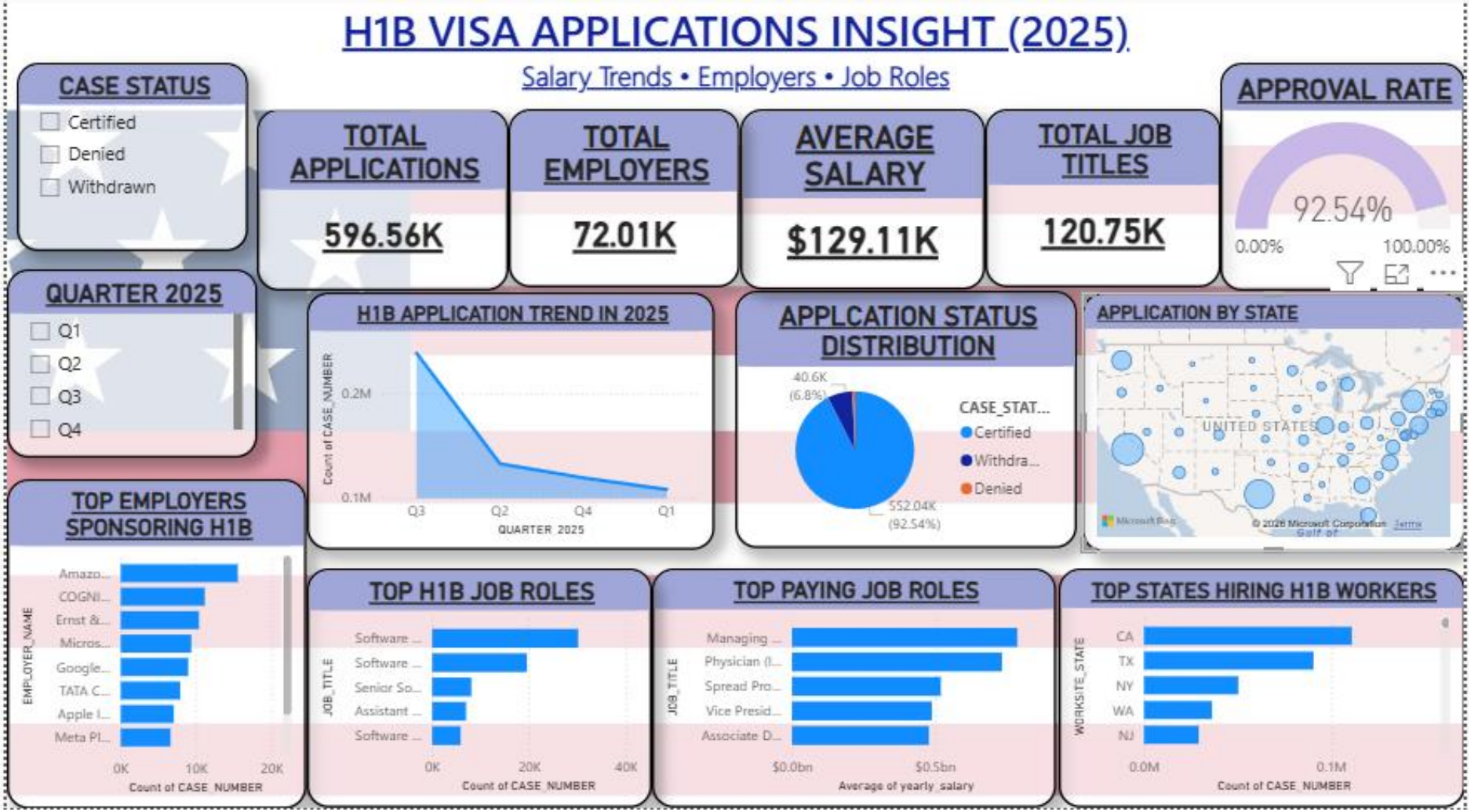
KEY INSIGHTS

- High Approval Rate (~92.5%) indicates **strong demand for skilled foreign workers.**
- Significant data reduction (2.4M → 596K) highlights the **importance of data cleaning and removing duplicates.**
- **Technology roles (e.g., Software Engineers) dominate H1B applications.**
- Top employers such as **Amazon, Google, and Microsoft** lead in visa sponsorship.
- **California, Texas, and New York** emerge as major hiring hubs.
- Salary distribution shows **higher pay in specialized and managerial roles.**

DASHBOARD OVERVIEW

- KPI Cards display key metrics like applications, salary, and approval rate
- Pie Chart shows distribution of application status – **Certified, Denied, Withdrawn**
- Line Chart highlights application trends across quarters
- Bar Charts represent top employers and job roles
- Map Visualization shows state-wise distribution of applications
- Filters (Case Status, Quarter) enable dynamic analysis

DASHBOARD


H1B VISA APPLICATIONS INSIGHT (2025)
Salary Trends • Employers • Job Roles
CASE STATUS
Certified
Denied
Withdrawn
TOTAL APPLICATIONS
596.56K
TOTAL EMPLOYERS
72.01K
AVERAGE SALARY
$129.11K
TOTAL JOB TITLES
120.75K
APPROVAL RATE
92.54%
0.00%
100.00%
QUARTER 2025
Q1
Q2
Q3
Q4
H1B APPLICATION TREND IN 2025
Count of CASE_NUMBER
0.2M
0.1M
Q3
Q2
Q4
Q1
QUARTER_2025
APPLCATION STATUS DISTRIBUTION
40.6K
(6.8%)
552.04K
(92.54%)
CASE_STAT...
Certified
Withdra...
Denied
APPLICATION BY STATE
UNITED STATES
TOP EMPLOYERS SPONSORING H1B
EMPLOYER_NAME
Amazo...
COGNI...
Ernst &...
Micros...
Google...
TATA C...
Apple I...
Meta Pl...
0K
10K
20K
Count of CASE_NUMBER
TOP H1B JOB ROLES
JOB_TITLE
Software ...
Software ...
Senior So...
Assistant ...
Software ...
0K
20K
40K
Count of CASE_NUMBER
TOP PAYING JOB ROLES
JOB_TITLE
Managing ...
Physician (I...
Spread Pro...
Vice Presid...
Associate D...
$0.0bn
$0.5bn
Average of yearly_salary
TOP STATES HIRING H1B WORKERS
WORKSITE_STATE
CA
TX
NY
WA
NJ
0.0M
0.1M
Count of CASE_NUMBER

Conclusion

- H1B visa applications show a consistently **high approval rate**
- **Technology roles dominate** the job market
- Major hiring is concentrated in key states like **California and Texas**
- Data-driven insights can help organizations improve hiring strategies

FUTURE WORK

THANK YOU